

INF-2700 Overview

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Database

- (large) collection of data
- related
- persistent
- shared
- personal, enterprise, global, web-enhanced
- +functionality

INF-2700 and other courses

- vs. INF-1101, Program = data structure + algorithm
 - data abstraction
 - large volume, persistent data
 - data sharing
 - reliability
- vs. INF-2201, operating systems
 - files, memory management
 - C programming

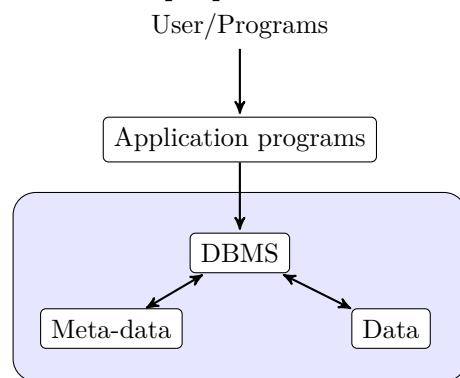
DBMS vs. file systems

- FS
 - low (no) data abstraction
 - program-data dependency
 - duplication of work: consistency, sharing, ...
 - flexible (UNIX philosophy)
- DBMS
 - addresses these issues
 - general purpose

History

- 60s, file processing
- 70s, hierarchical, network
- 80s, relational
- 90s, client/server, OO
- now, distributed, parallel, web, XML, NoSQL, data mining

Database and people



People

- end user
- database administrator
- database designer
- system analyst
- application developer
- DBMS system/tool designer/developer

Three aspects of INF-2700

1. modeling and design of databases (db designer, sys analyst)
2. DB programming (app. developer)
3. DBMS implementation (DBMS designer/developer) > 50% of the course

Textbook

Database System Concepts, 7th Edition, A. Silberschatz, H. Korth and S. Sudarshan, McGraw Hill, 2019 ISBN 978-1-260-08450-4

Assignments

- 3 programming assignments
- relational database, DBMS elements
- C, SQL

Colloquiums

- in the PC lab
- will work on earlier exam questions